IFRS 9 Expected Credit Loss Engine with an Independent Model-Challenge Layer

Project Report

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1. Introduction

This report documents an end-to-end credit-risk project built around the IFRS 9 standard for loss provisioning. The deliverable is a working engine that computes the Expected Credit Loss (ECL) of a loan portfolio, covering every step from the probability-of-default model through to committee-level reporting, together with the independent validation layer that challenges it. The project was built on real public data from Lending Club, is fully reproducible in a notebook, and is accompanied by an interactive dashboard.

Its purpose extends beyond the production of a provisioning figure. It sets out to demonstrate the two complementary postures of a modern bank risk function: the first line that produces the figure, and the second line of defence that independently contests it.

2. Context

Since its entry into force, IFRS 9 has required banks to provision for expected credit losses rather than for incurred losses alone. Provisions must be forward-looking, must incorporate macroeconomic scenarios, and must classify each exposure into one of three stages reflecting how its credit risk has evolved since origination. In parallel, the Basel prudential framework governs regulatory capital through internal models whose outputs an institution must be able to measure, explain and challenge.

Within this framework, the risk function operates as an independent control. It neither takes positions nor owns the front-office models. Instead, it validates them by assessing their discrimination, calibration and stability, and it determines what the institution can rely upon. The project was designed to reproduce that environment faithfully on a realistic dataset.

3. Problem statement

How can a bank estimate, provision and stress the expected credit losses of a loan portfolio under IFRS 9, while retaining a model that is at once performant and auditable, and challenging that model independently as a second line of defence would?

Answering this question requires reconciling two tensions that lie at the heart of the role. The first opposes performance to interpretability in the choice of model. The second opposes the production of a figure to its independent validation.

4. Scope of the work

A complete ECL pipeline was built and run on real data. It covers every component that a provisioning exercise requires:

- a probability-of-default (PD) model in two versions, an interpretable logistic scorecard serving as the official model and a gradient-boosting challenger;
- Loss Given Default (LGD) and Exposure At Default (EAD) parameters, each supported by documented assumptions;
- an ECL engine implementing IFRS 9 staging (Stage 1, 2 and 3), the Significant Increase in Credit Risk (SICR) logic, and the loss computation;
- a forward-looking macroeconomic overlay weighting baseline, adverse and upside scenarios;
- an independent validation layer covering discrimination, calibration, stability and backtesting;
- explainability through SHAP, and an interactive Streamlit dashboard for reporting.

5. Methodology

5.1 Data and default definition

The analysis uses a real Lending Club loan sample of 21,668 resolved loans. Default was fixed at the outset as charge-off or default and held constant throughout. Only resolved loans, whether fully paid or defaulted, were retained, so that the target is genuinely observed. The data was split by time, training on earlier vintages and testing on later ones, in order to reproduce a true out-of-time evaluation rather than an optimistic random split.

5.2 Probability-of-default modelling

Two models were trained on the same origination-time features, including the interest rate, loan amount, FICO score, debt-to-income ratio, income and employment length. The logistic scorecard serves as the official and interpretable model, while the gradient-boosting challenger quantifies the maximum performance attainable. Both were calibrated so that their predicted scores can be read as genuine probabilities.

5.3 ECL engine, staging and SICR

Each exposure is classified as Stage 3 when it is in default, as Stage 2 when a significant increase in credit risk is detected relative to origination, and as Stage 1 otherwise. The Stage 2 trigger combines a relative test, an absolute test and a thirty-days-past-due backstop. The loss is computed as ECL = PD × LGD × EAD, using a twelve-month PD in Stage 1 and a lifetime PD in Stages 2 and 3. The SICR threshold is exposed as a parameter, so that its effect on provisions can be demonstrated directly.

5.4 Forward-looking overlay and validation

Probabilities of default are scaled across three macroeconomic scenarios, and the reported ECL is their probability-weighted average. The validation layer then computes discrimination (AUC, Gini and KS), calibration (predicted against observed default rates by grade), stability (the Population Stability Index) and a backtest, reproducing the battery of tests that a model-risk reviewer would run.

6. Results

All figures below are computed on the out-of-time test portfolio.

6.1 Probability-of-default performance

Model	AUC	Gini	KS
Logistic scorecard (official)	0.681	0.361	0.273
Gradient-boosting challenger	0.684	—	—

The challenger improves on the interpretable scorecard by only three thousandths of AUC, an immaterial gain. The scorecard is therefore retained as the official model, a decision taken on governance grounds rather than on raw performance.

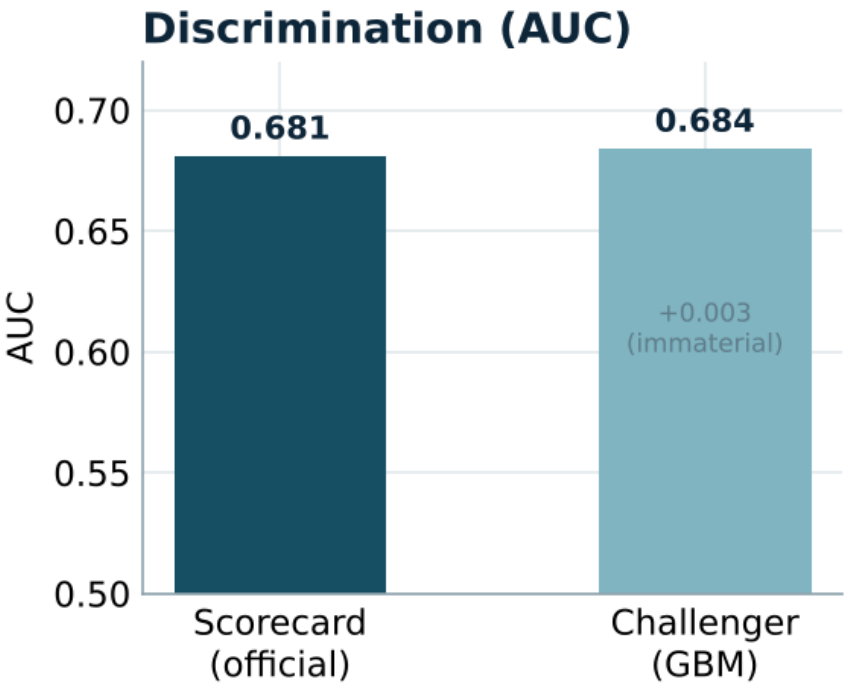


Figure 1. Discrimination of the two models. The challenger's advantage is negligible.

6.2 Calibration and backtest by grade

Grade	PD predicted	Default rate observed	Gap
A	0.100	0.086	+0.014
B	0.199	0.197	+0.002
C	0.299	0.302	−0.003
D	0.393	0.375	+0.017
E	0.446	0.440	+0.006

Predicted and observed default rates remain close and increase monotonically from grade A to grade E, which indicates a well-calibrated model across the bulk of the portfolio. The gaps widen only for the sparse F and G grades, where the number of observations is small.

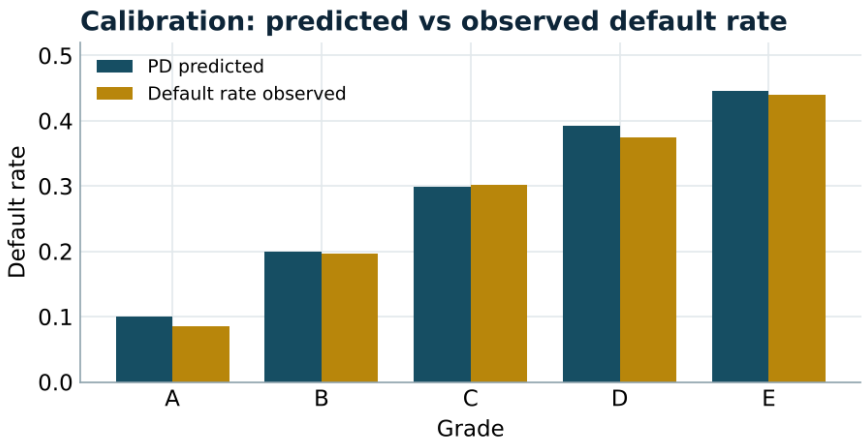


Figure 2. Predicted and observed default rates track each other closely across grades.

6.3 Expected Credit Loss by stage

Stage	Exposures	EAD	ECL	Coverage
Stage 1	5,557	67.1M	6.9M	10%
Stage 2	2,518	45.3M	17.7M	39%
Stage 3	2,874	46.8M	21.1M	45%
Total	**10,949**	**159.2M**	**45.6M**	**29%**

Coverage rises sharply across the stages, from 10 per cent to 39 per cent and then 45 per cent. This is the expected pattern, since Stages 2 and 3 carry lifetime losses and therefore concentrate the bulk of provisions on a minority of exposures.

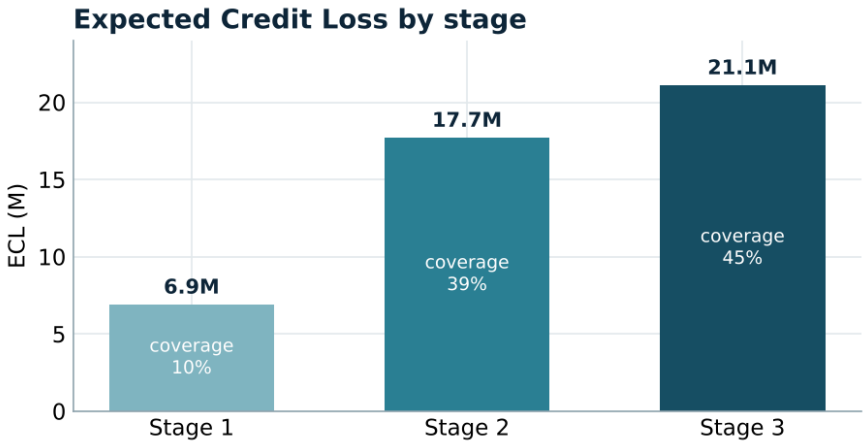


Figure 3. Expected credit loss and coverage ratio by IFRS 9 stage.

6.4 Forward-looking sensitivity

Scenario	Weight	ECL
Baseline	0.5	45.6M
Adverse	0.3	61.7M
Upside	0.2	35.3M

Probability-weighted	**1.0**	**48.4M**

A stressed macroeconomic path lifts provisions by 35 per cent relative to baseline, which is precisely the sensitivity that a provisioning committee needs to monitor.

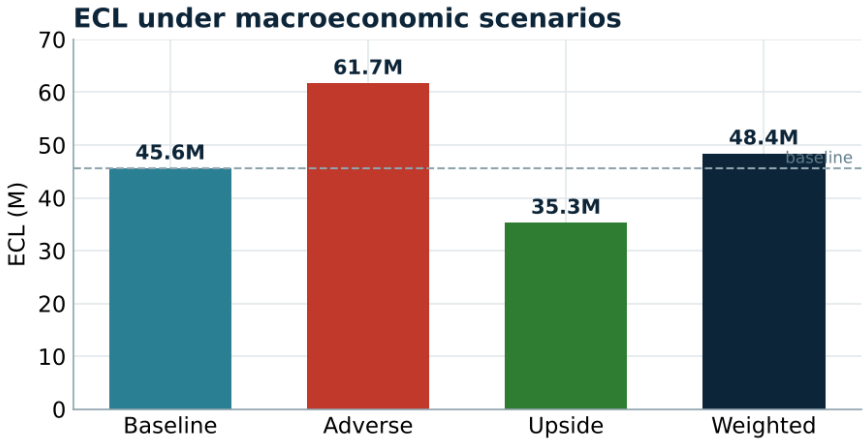


Figure 4. Expected credit loss under baseline, adverse and upside scenarios, with the probability-weighted total.

6.5 Explainability

A SHAP analysis of the challenger confirms that the model relies on economically sensible drivers, namely the interest rate, loan amount, FICO score and debt-to-income ratio, each acting in the expected direction. This renders the model auditable, which is a precondition for its use in a controlled environment.

> **Headline result.** On real Lending Club data, the pipeline produces a total ECL of 45.6M at 29 per cent coverage, rising to 61.7M under an adverse scenario. The figure is derived from a calibrated scorecard with an AUC of 0.68, retained over a marginally stronger but opaque challenger, and every result is independently validated.

7. Conclusion

Returning to the problem statement, the project demonstrates that a bank can estimate, provision and stress IFRS 9 expected credit losses from end to end while retaining a model that is both performant and auditable. The central tension is resolved explicitly. The interpretable scorecard was preferred to the opaque challenger because the performance difference proved immaterial, and the challenger's reliance on sensible drivers was verified through SHAP rather than assumed.

Equally important, the provisioning figure is not produced in isolation. It is accompanied by an independent validation covering discrimination, calibration, stability and backtesting, as well as a scenario-based sensitivity analysis. The system therefore embodies the second-line-of-defence posture in concrete form, since it builds both the number and its contestation.

8. Limitations and next steps

- The public sample contains no recovery data, so LGD relies on a documented 45 per cent unsecured assumption. The full Lending Club file would permit an observed LGD.

- Lifetime PD is approximated from the twelve-month PD. A term-structure model would provide the production refinement.
- A bridge to the Basel IRB Expected Loss on the same portfolio would allow the gap between accounting and regulatory losses to be shown directly.

Stating these limitations explicitly is itself a second-line posture, since a controller sets out what a model cannot do rather than concealing it. The full code, tests and dashboard are available at github.com/ArjiNasim/ifrs9-ecl-engine.